

Healthcare Data Analysis: Patient, Treatment, and Cost Insights

Exploratory Data Analysis and Prescription Text Processing

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Agenda

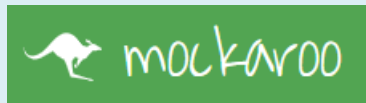
- Project Motivation & Goals
- Datasets Overview
- Methodology
 - Exploratory Data Analysis (EDA)
 - Unstructured Text Processing
- Key Findings
- Conclusions
- Future Work

Project Motivation

- Healthcare systems generate large amounts of patient and treatment data
- Understanding patterns can help:
 - Improve resource allocation
 - Identify cost drivers
 - Analyze treatment effectiveness

**WHY THIS
PROJECT?**

- **1st Objective: Dataset for analysis**



<https://www.kaggle.com/>

Project Goal

- Extract meaningful insights about COSTS from healthcare data
- Identify cost drivers



Datasets Overview

<https://www.kaggle.com/>

HealthcareDataset1 (Patients info)

- PatientID,
- PatientName
- Age
- Gender
- Bloodtype
- Diagnosis
- Treatment
- Admission Date
- Discharge Date
- Total Bill (\$)
- Full Prescription Details
(unstructured)

1000 rows, 11 columns

HealthcareDataset2 (Hospital info)

- PatientID
- Hospital
- DoctorName
- Room Number
- Daily Cost (\$)
- TreatmentType
- RecoveryRating

1000 rows, 7 columns

Methodology

Data Analytics Pipeline



Exploratory Data Analysis

Data Quality Check, Cleaning & Data Transformation

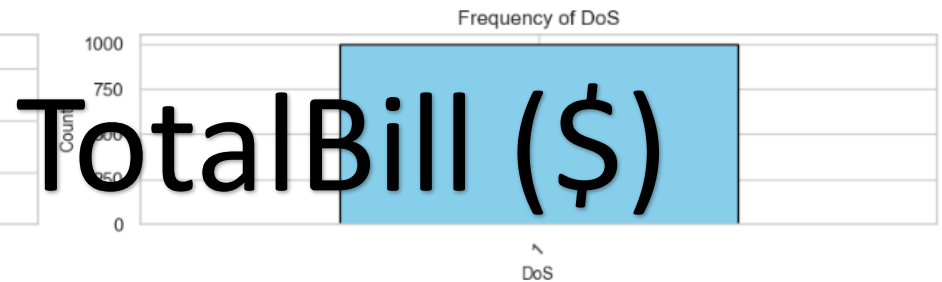
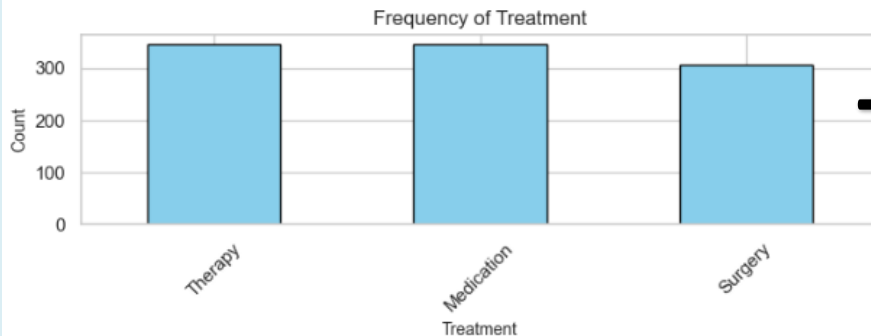
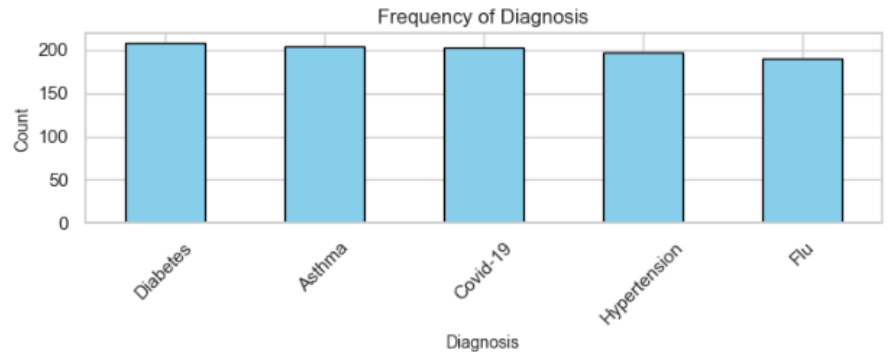
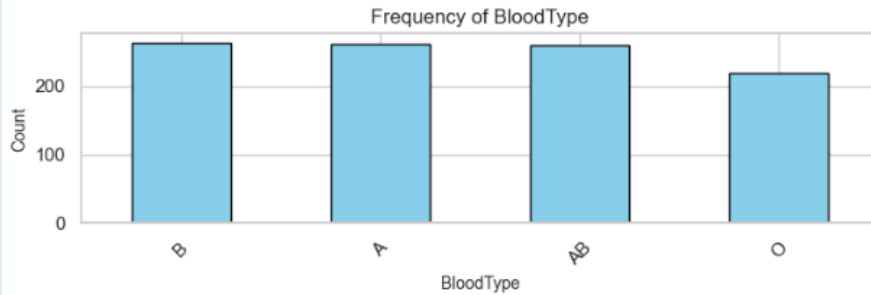
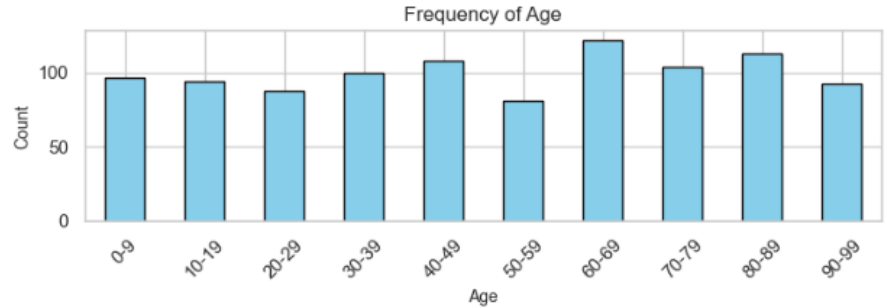
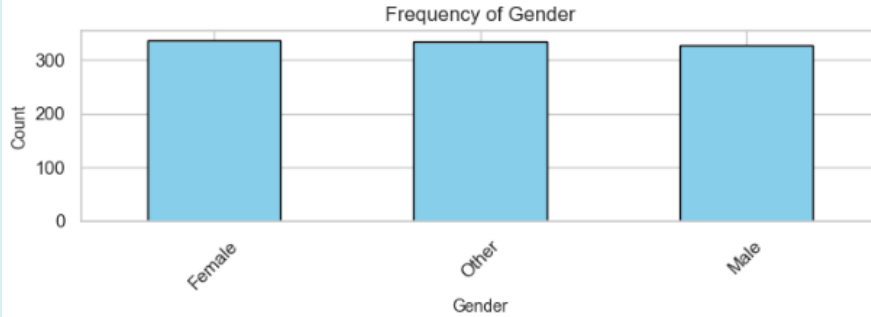
- Date consistencies
 - Identify Nulls
 - Patients (6% TotalBill)
 - Doctors (7,8% RecoveryRate)
 - Identify Duplicates
 - Identify Outliers
 - Extract information from the “Full Prescription Details” (patients)
 - Combined dataset (PatientID):
- DoS (Days of Stay)
 - Rh (+/-)
 - Temporary Analysis
 - AdmissionDate
 - AD_Year
 - AD_Quarter
 - AD_Month
 - AD_Day
 - DischargeDate
 - DD_Year
 - DD_Quarter
 - DD_Month
 - DD_Day



df = clinical information + hospital operational data (# 1000 rows, 32 variables)

Patients Breakdown

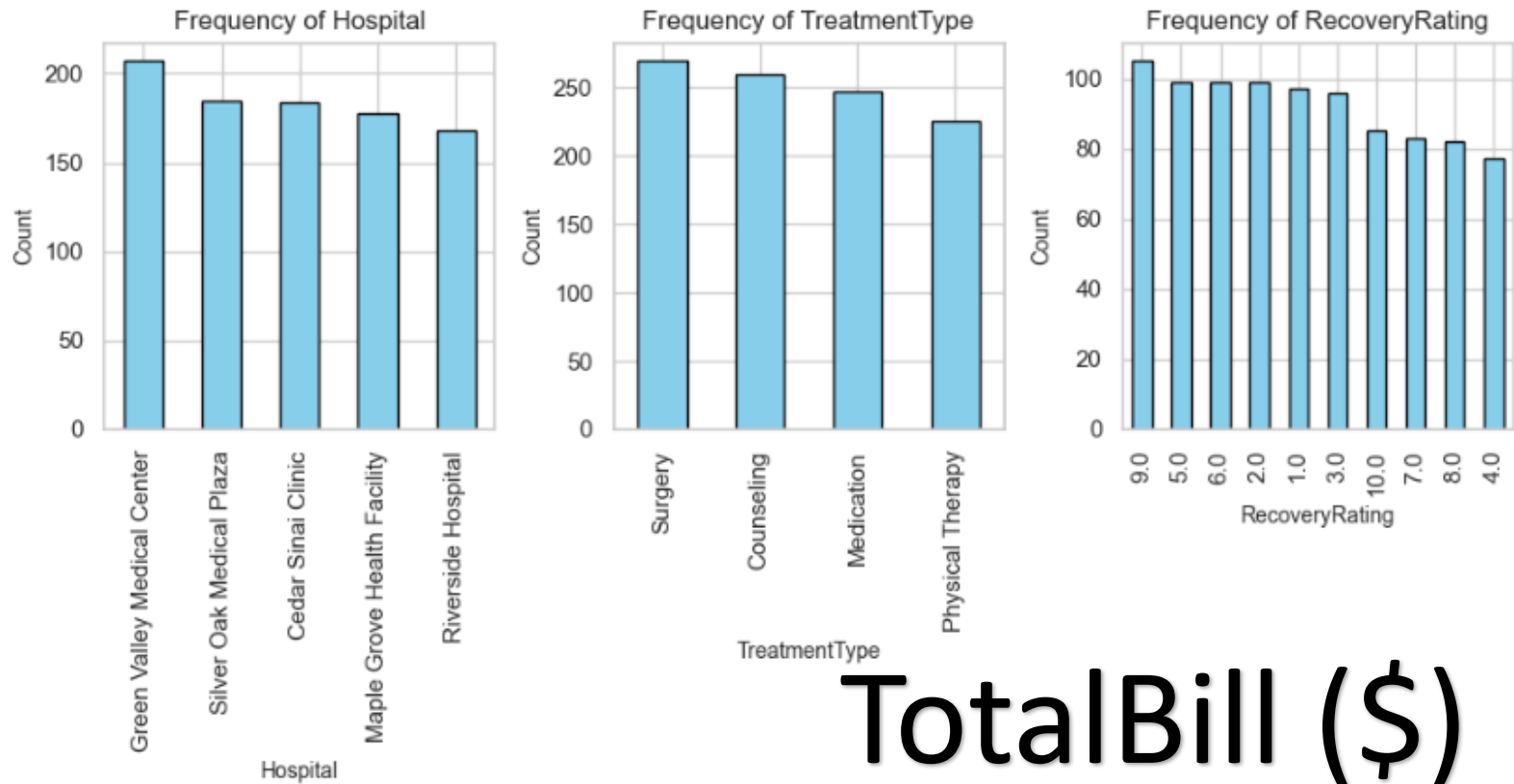
Frequency Distribution of Patients Data



Total Bill (\$)

Hospital Breakdown

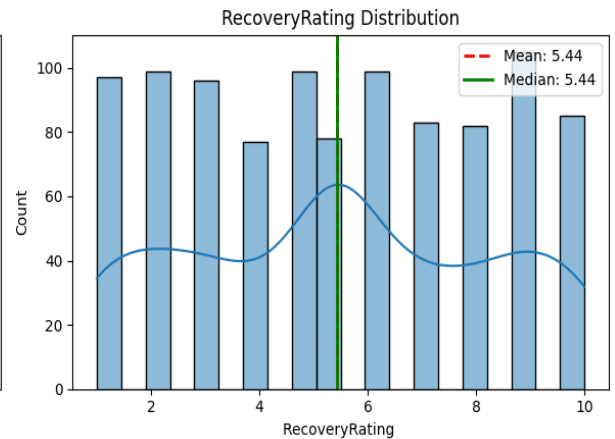
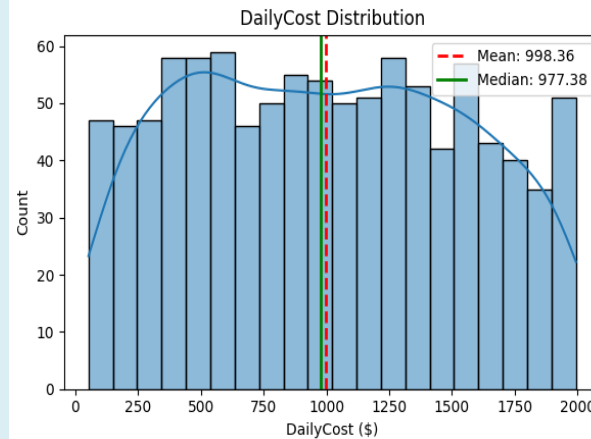
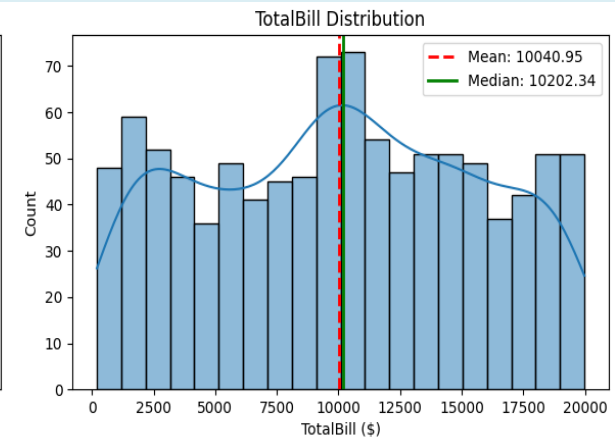
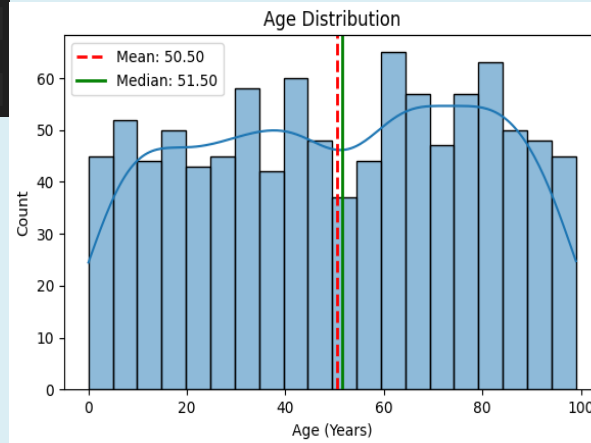
Frequency Distribution of Hospital Data



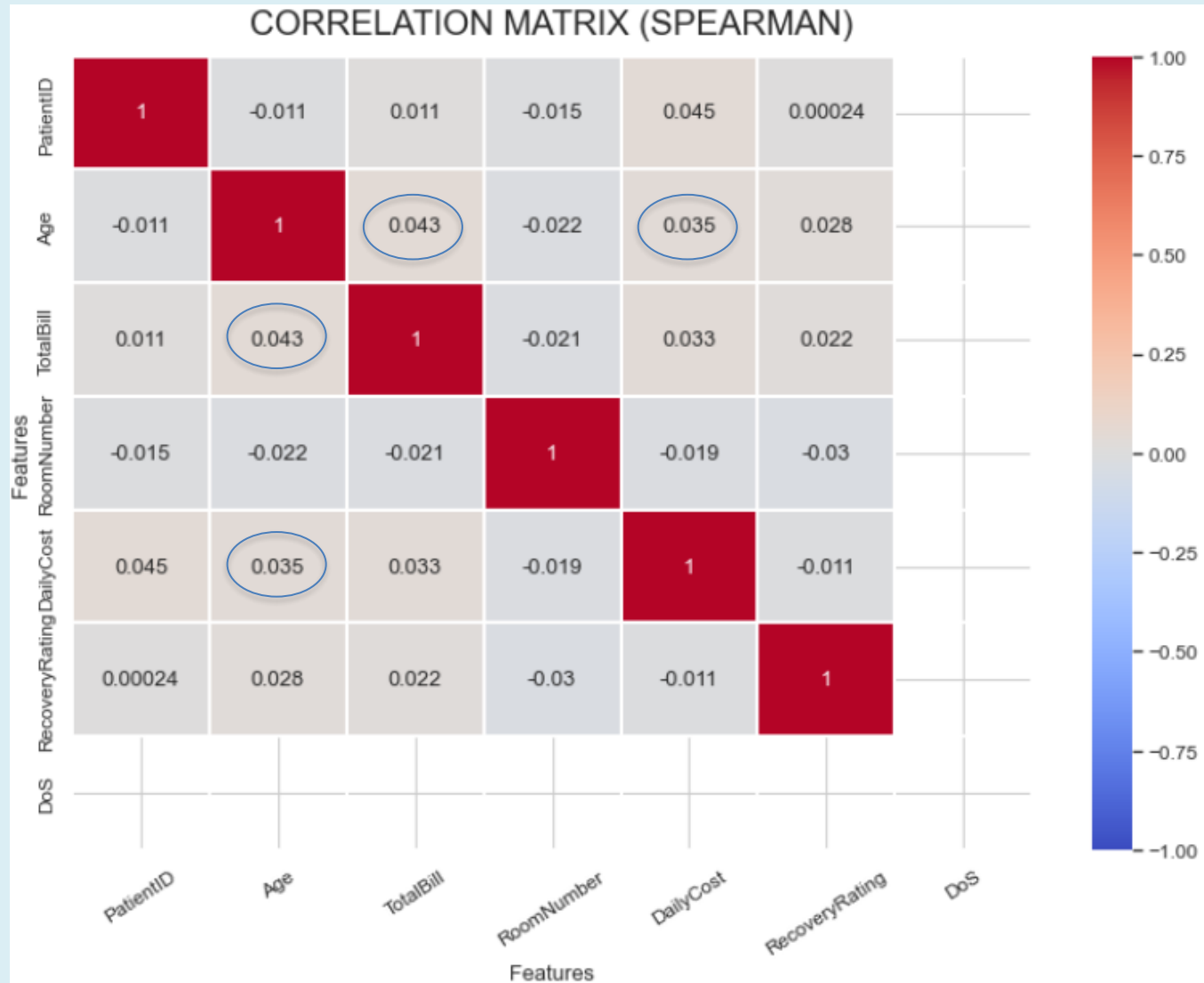
Descriptive Statistics & Variable Distribution

	Age	TotalBill	DailyCost	RecoveryRating	DoS
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.0
mean	50.500000	10040.947498	998.358349	5.436009	1.0
std	28.599859	5626.658910	545.178933	2.776767	0.0
min	0.000000	200.928022	52.023698	1.000000	1.0
25%	26.000000	5326.999625	527.573516	3.000000	1.0
50%	51.500000	10202.339277	977.376355	5.436009	1.0
75%	75.000000	14552.941322	1459.068550	8.000000	1.0
max	99.000000	19979.201527	1994.047594	10.000000	1.0

Normal Distributions???



Quantitative Variables Correlation



Statistical Test Results

Significant Differences and Associations

(T-test and Chi²)

Division variable: Age

T-statistic: -2.280, p-value: 0.023

→ There is a statistically significant difference in 'TotalBill' between the two groups based on the median of the variable.

Division variable: DailyCost

T-statistic: -0.828, p-value: 0.408

→ No significant difference in 'TotalBill' between the groups based on the median of the variable.

Division variable: RecoveryRating

T-statistic: -0.147, p-value: 0.883

→ No significant difference in 'TotalBill' between the groups based on the median of the variable.

Variable: Gender

Chi2: 1915.7667, p-value: 0.4330

→ No significant relationship with TotalBill

Variable: BloodType

Chi2: 2860.0548, p-value: 0.4909

→ No significant relationship with TotalBill

Variable: Diagnosis

Chi2: 4000.0000, p-value: 0.0167

→ There is a significant relationship with TotalBill

Variable: Treatment

Chi2: 2000.0000, p-value: 0.0657

→ No significant relationship with TotalBill

Variable: Hospital

Chi2: 3527.6197, p-value: 0.4986

→ No significant relationship with TotalBill

Variable: DoctorName

Chi2: 350587.0304, p-value: 0.5552

→ No significant relationship with TotalBill

Variable: TreatmentType

Chi2: 2863.2094, p-value: 0.4743

→ No significant relationship with TotalBill

Variable: Medicine1

Chi2: 27663.8823, p-value: 0.4534

→ No significant relationship with TotalBill

Variable: Medicine2

Chi2: 27652.0842, p-value: 0.4733

→ No significant relationship with TotalBill

Variable: Medicine3

Chi2: 18929.1735, p-value: 0.4553

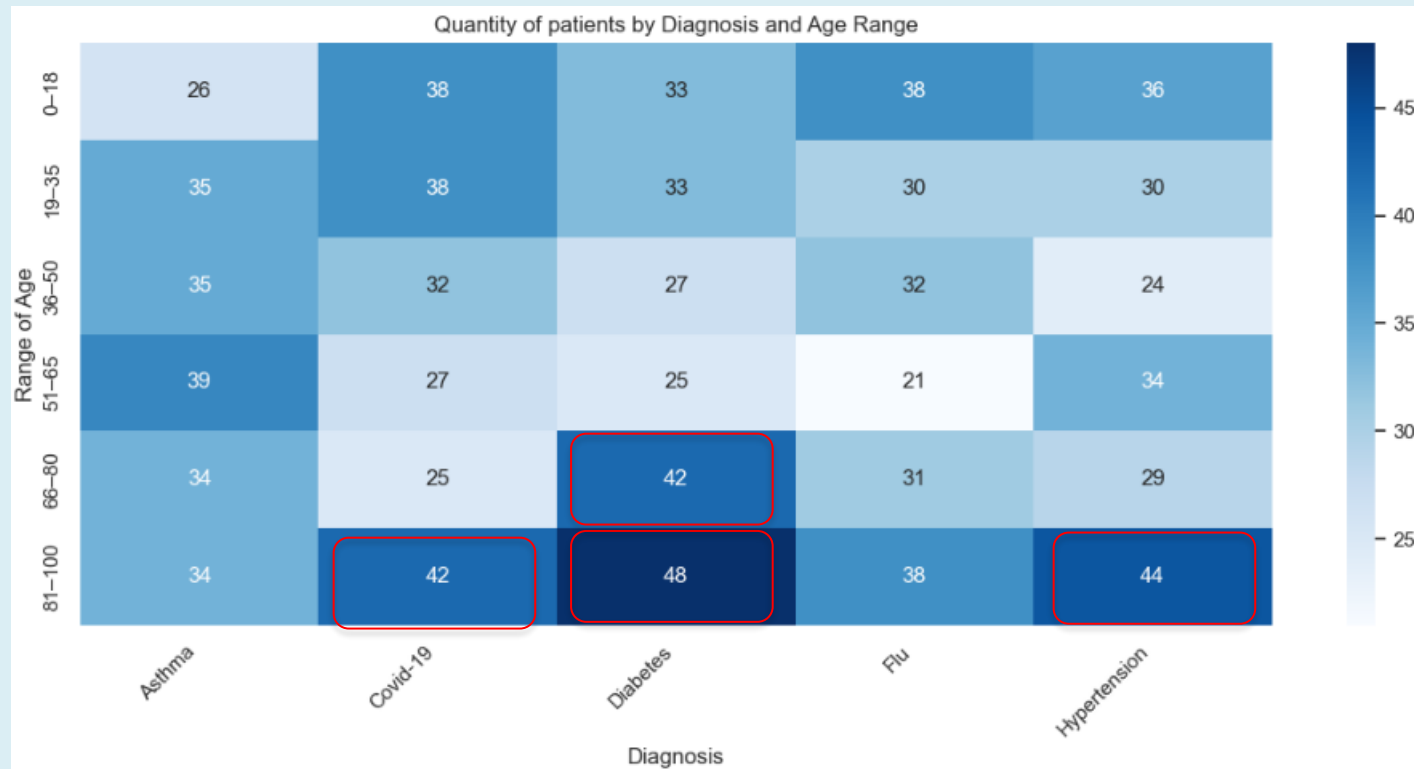
→ No significant relationship with TotalBill

Variable: Medicine4

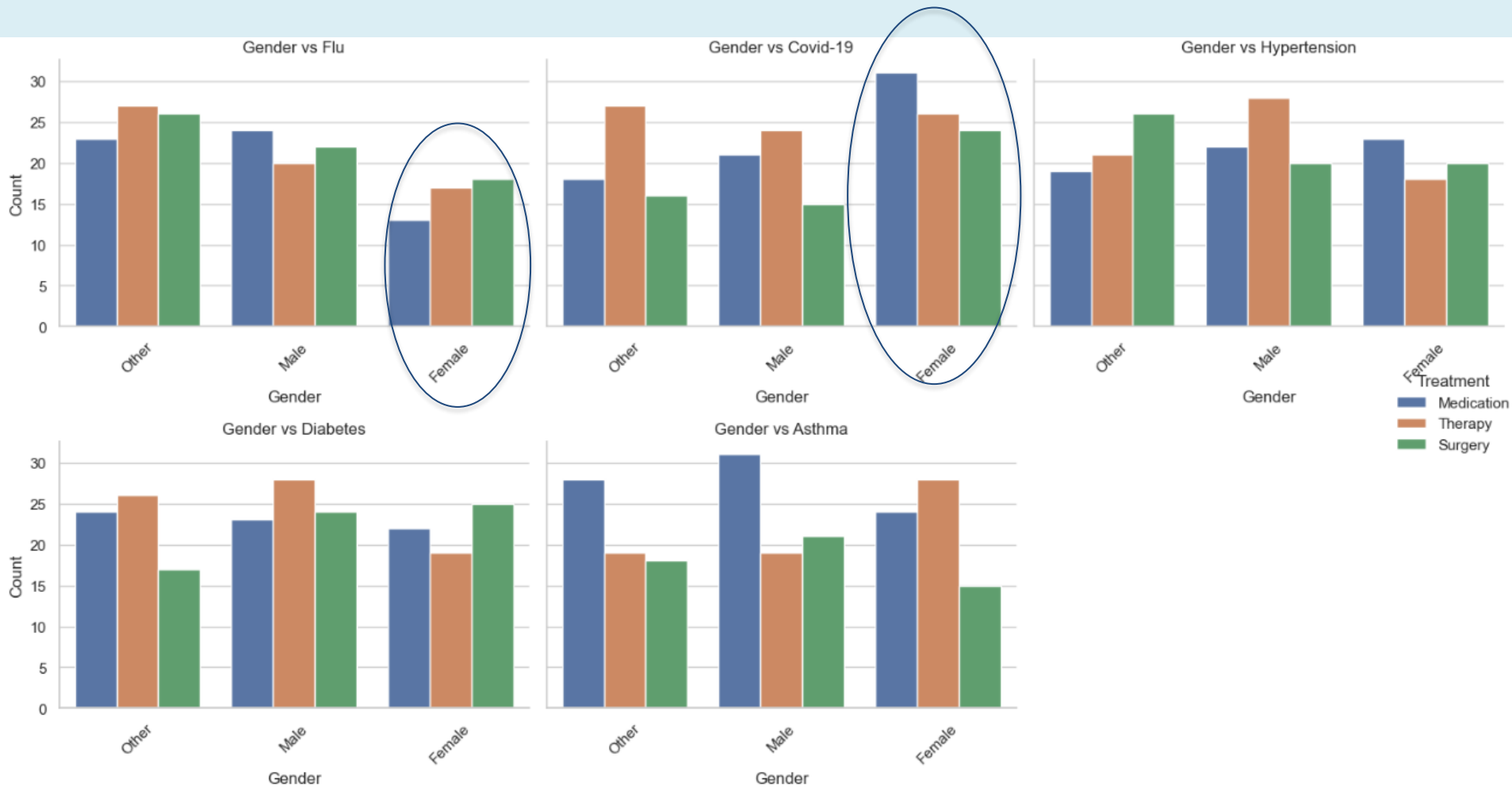
Chi2: 10147.4854, p-value: 0.4244

→ No significant relationship with TotalBill

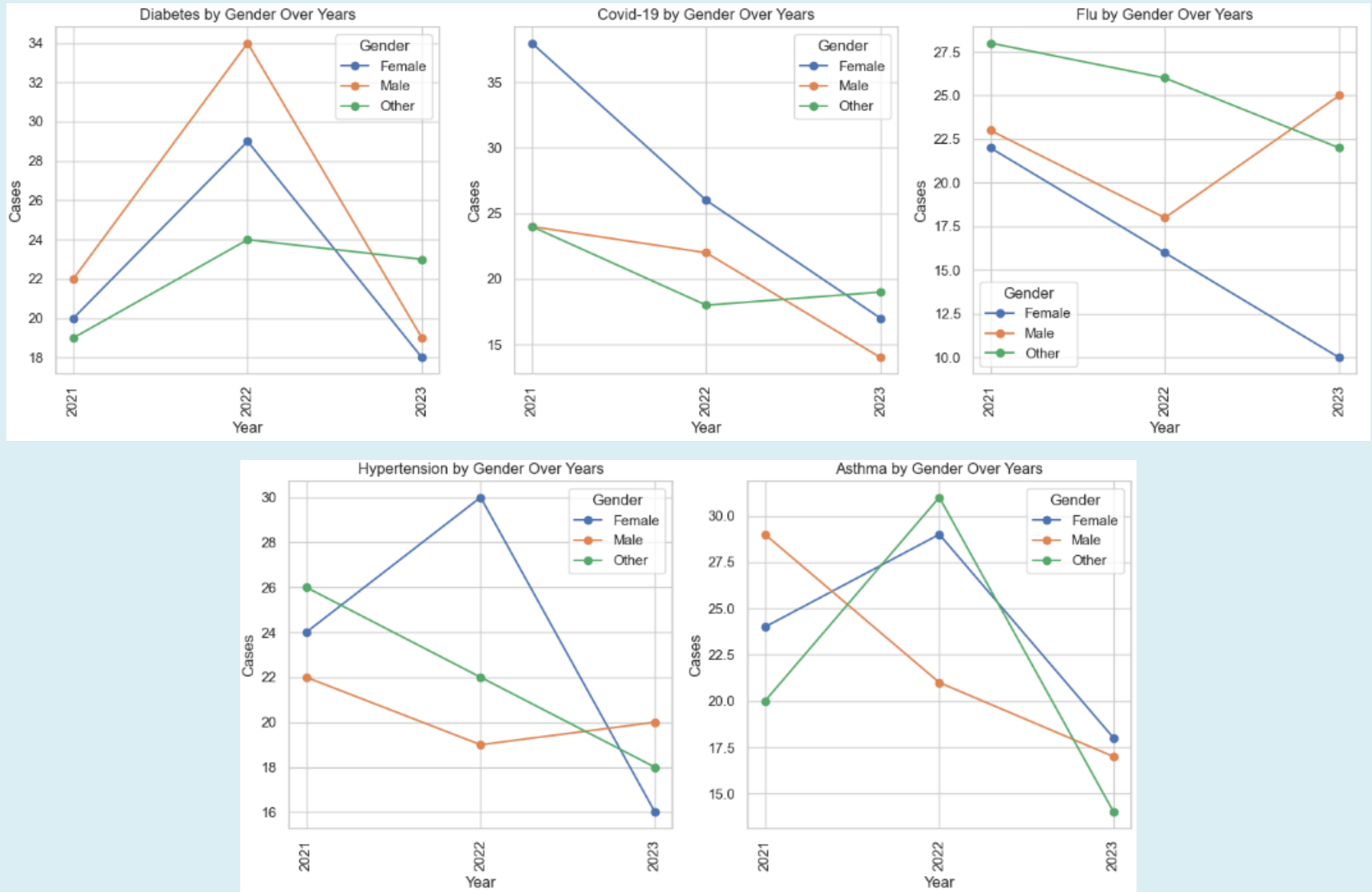
Key Findings: Diagnosis per Age



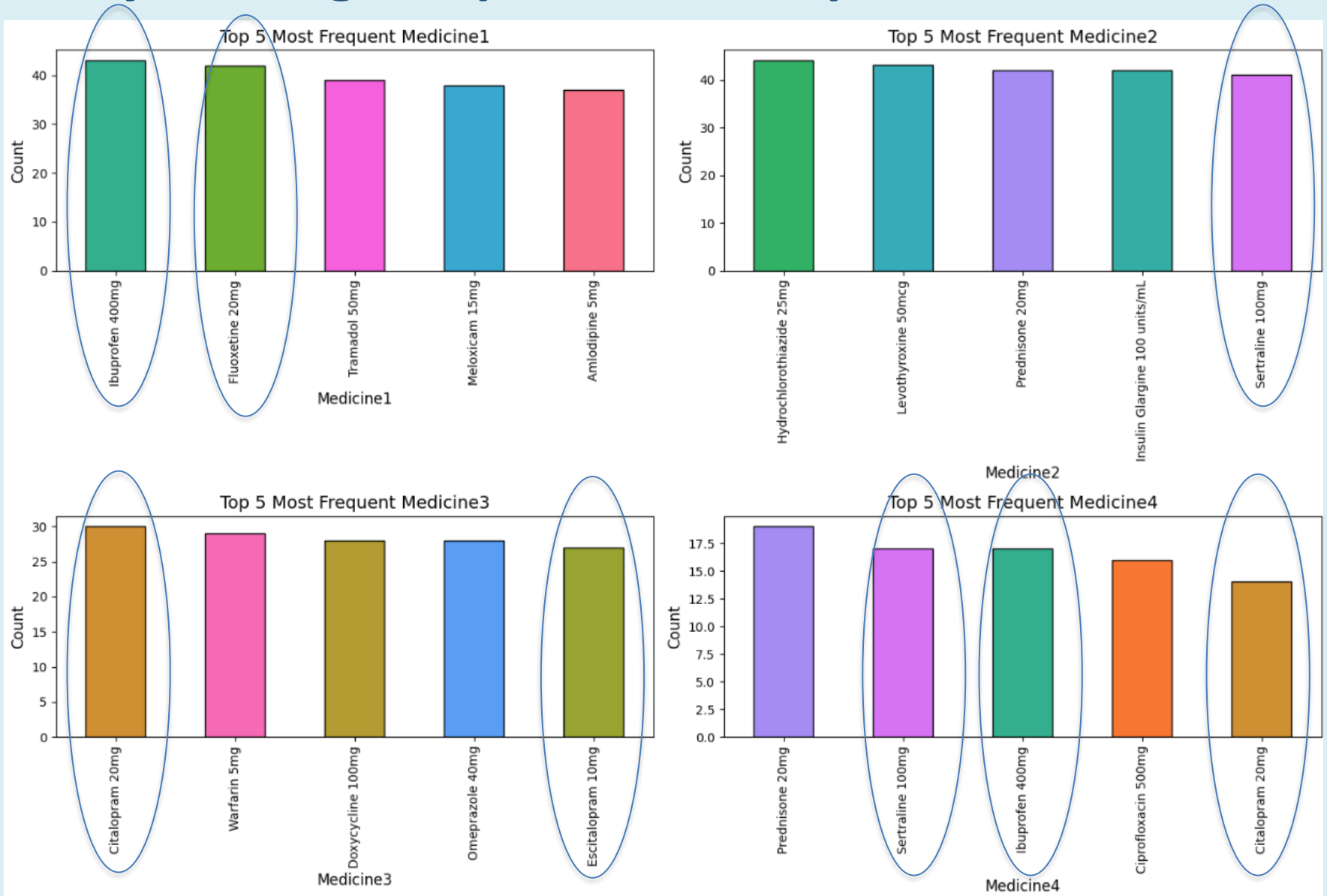
Key Findings: Diagnosis per Gender



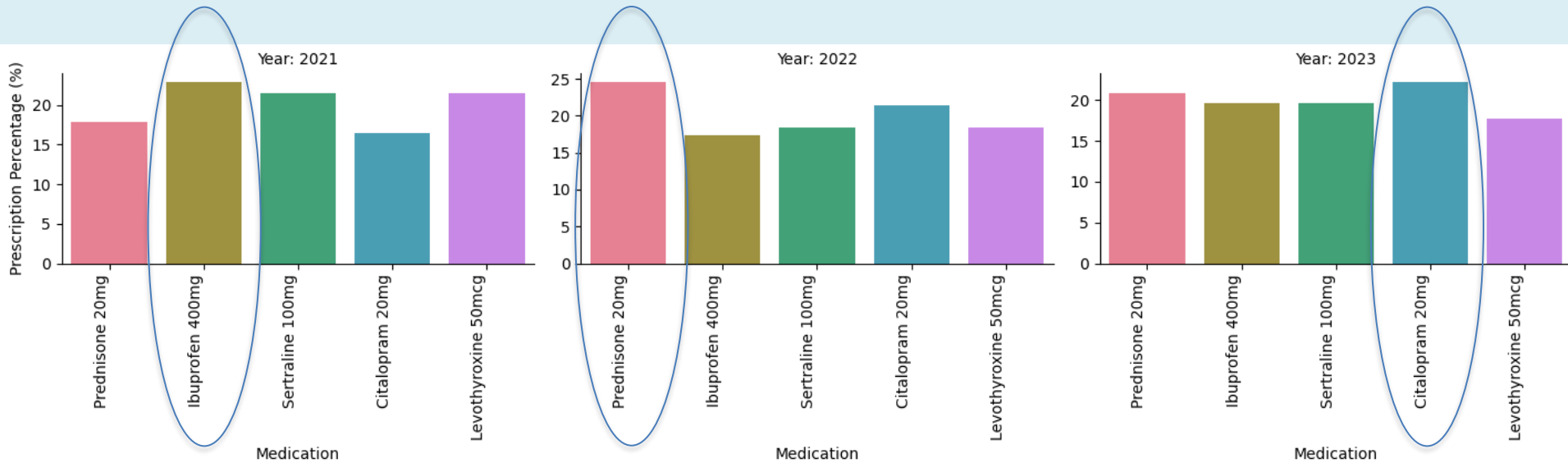
Key Findings: Diagnosis by gender over years



Key Findings: Top 5 Most Frequent Medicines



Key Findings: Top 5 Most Frequent Medecines (over years)

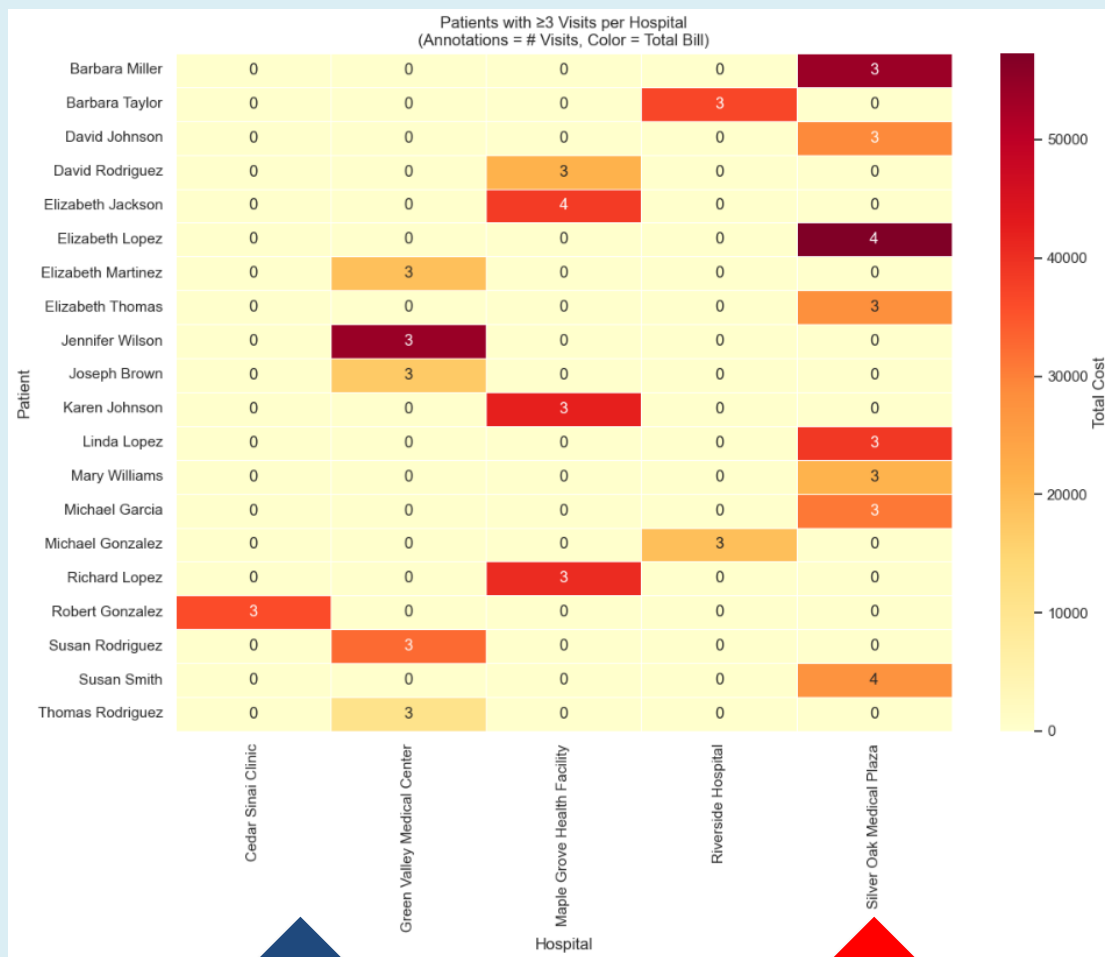


YEAR 2021 – Ibuprofen 400 mg

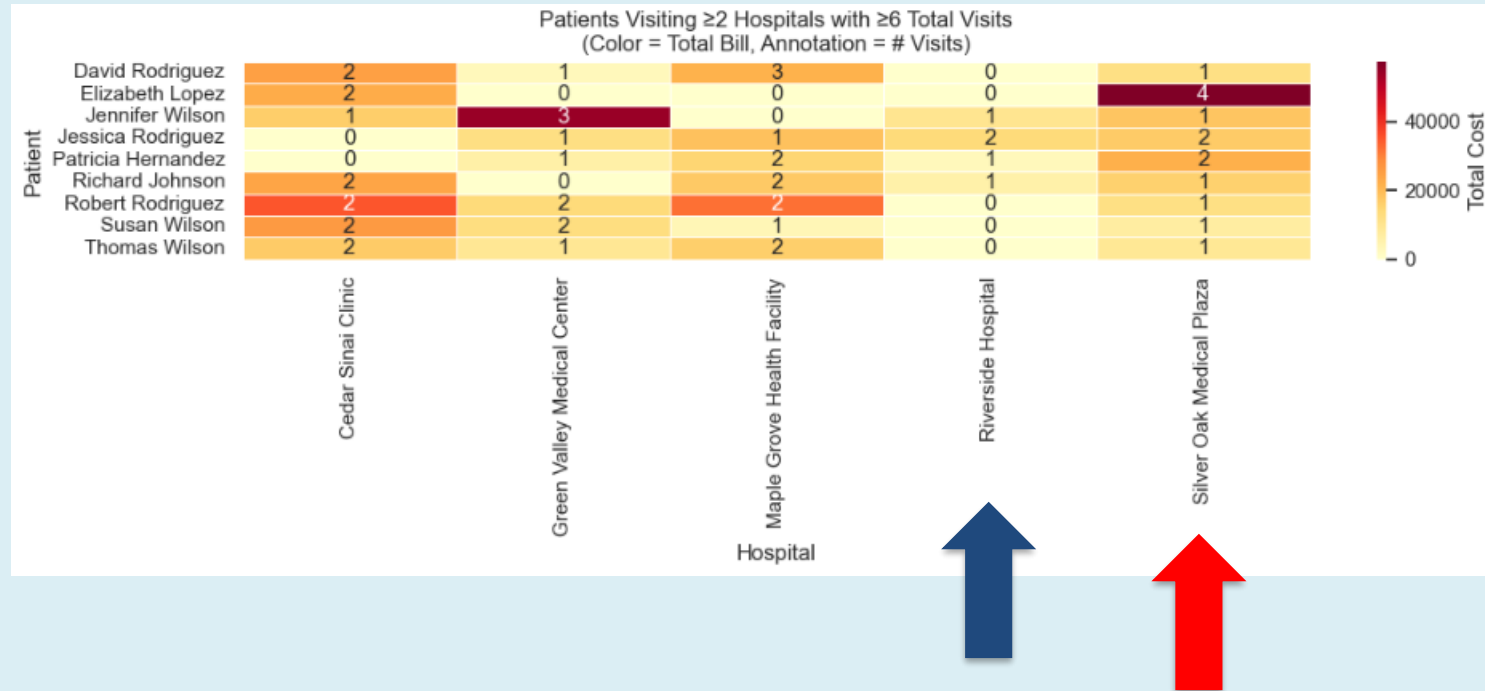
YEAR 2022 – Prednisone 20 mg

YEAR 2023 – Citalopram 20 mg

Key Findings: Patients with recurrence (I)



Key Findings: Patients with recurrence (II)



David Rodriguez – 7 visits in 4 hospitals

Robert Rodriguez – 7 visits in 4 hospitals

Conclusions

- Successful Data Cleaning & Transformation
- Basic Text Information Extraction (“Full prescription Details”)
- Successful Exploratory Data Analysis
- Poor Identification of Clinical vs Economic Insights
 - Most variables do not show significant relationship with cost
 - Only Age and Diagnosis exhibit statistically relevant differences
 - Elderly patients with chronic diseases account for the majority of hospital visits
 - Preliminary analysis shows a decline in hospital visits in 2023
 - Steady growth observed in mental health medication prescriptions
 - Patient repeat visits vary by hospital
- Main limitation is the synthetic nature of dataset

Future Work

- Apply Machine Learning Models to:
 - Predict TotalBill (cost)
 - Predict RecoveryRating
- Use more advanced NLP techniques for
 - Prescription analysis
- Include additional variables:
 - Comorbidities (ie. Diabetes, Hypertension)
 - Laboratory results (i.e glucose, creatinine...)
- Validate findings on real-world healthcare datasets

Thank you

"You lose your curiosity when you stop learning"

Katherine Johnson- Physicist, Space Scientist and Mathematician